Dynamic Set-Theoretic Normalization of Provenance-Bearing Hyper-Relational Frames

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Abstract

We develop a deterministic semantics for normalizing claims and event frames across a mutable document corpus. The extraction front end may propose several frame interpretations for each claim occurrence, and coreference is not forced to one early clustering. Instead, every exact interpretation and strict coreference partition is a candidate normalization hypothesis. Constraints denote property sets of hypotheses, and the solver is literally set theoretic estimation (STE): it repeatedly intersects its current state with property sets and records the exact set reduction. The resulting feasible family supports a possible-worlds semantics for identity. Coreference holding in every feasible hypothesis defines a safe quotient; disagreement between feasible merge and split hypotheses defines an explicit uncertainty overlay U.

We prove document-insertion antitonicity, deletion expansion, insertion order independence, a delta decomposition by newly activated provenance constraints, identity and composition laws for feasible-world restriction, equivalence of must-coreference, and exact status separation into inconsistent, must, cannot, and uncertain cases. A finite countermodel proves that possible coreference and the uncertainty overlay need not be transitive even though every exact hypothesis is a partition. We prove existence of finite minimal document supports for must- and cannot-coreference conclusions. Finally, we lift a downstream truth feasible set over every normalization world and prove monotonicity and a commutation theorem for normalization-independent truth semantics. The accompanying Lean 4 development names every stated formal result.

1 Introduction

Cross-document event coreference is usually presented as clustering: a learned model scores pairs or clusters, and an agglomerative or graph algorithm selects one partition. This is an effective predictive architecture, but it is an awkward database semantics. A database receives and retracts documents. A new source can distinguish events previously thought identical; deletion can remove the only evidence that forced a merge or split. Anaphora and omitted arguments can remain genuinely unresolved. A destructive union operation has neither the provenance nor the inverse required by this setting.

Event identity itself is richer than lexical similarity. Work on quasi-identity and dense event annotation distinguishes strict identity from subevent, near-identity, and related-event phenomena [5, 7, 8]. Recent resources make decontextualization, difficult identity boundaries, and framing divergence increasingly explicit [9, 10]. The correct deterministic reaction is not to pretend every ambiguity has a unique answer. It is to retain the admissible answers and state exactly what follows in all, some, or none of them.

This paper formalizes that reaction. We assume a high-recall extraction front end. Each immutable claim occurrence has source provenance and a nonempty set of candidate hyper-relational frames. An exact normalization hypothesis chooses one candidate per claim and supplies an equivalence relation of strict frame identity. Typed nonidentity links—subevent, causation, temporal relation, repeated instance, framing divergence—are conceptually separate and do not participate in the identity quotient.

The contributions are:

1. an STE solver semantics in which property-set application and exact reduction are the complete operational core;
2. a provenance-indexed corpus semantics supporting insertion and deletion without destructive clustering;
3. a must-coreference quotient and a nontransitive uncertainty overlay;
4. a finite countermodel ruling out possible-coreference clustering;
5. existence of minimal finite document certificates for settled links;
6. contravariant feasible-world restriction and covariant canonical-frame maps, exposing the fundamental insertion/deletion asymmetry;
7. a dependent lift of downstream truth estimation over all feasible normalizations.

2 The normalization universe

Fix types of documents, claim occurrences, structured frames, exact normalization hypotheses, and constraints:

$$\mathsf{D}, \qquad \mathsf{C}, \qquad \mathsf{F}, \qquad \Omega, \qquad \mathsf{K}.$$

The Lean structure implementing these data is `STE.DynamicFrame.Model`.

Each claim c has an immutable source document $\mathrm{owner}(c)$ and a nonempty candidate set $\Phi(c) \subseteq \mathsf{F}$. Each exact hypothesis $\omega \in \Omega$ chooses

$$\mathrm{interpret}_\omega(c) \in \Phi(c)$$

and supplies a strict-coreference relation $\sim_\omega$ on claims. The model requires $\sim_\omega$ to be an equivalence relation. Thus every exact hypothesis is an ordinary, logically valid partition. Ambiguity is represented between hypotheses, not by allowing one hypothesis to contain an invalid overlapping "partition."

Every constraint $k \in \mathsf{K}$ has two denotations. First, its provenance support is a set of documents

$$\mathrm{supp}(k) \subseteq \mathsf{D}.$$

Second, it denotes an STE property set

$$S_k = \{\omega \in \Omega \mid \omega \text{ satisfies } k\}.$$

For an active corpus $D \subseteq \mathsf{D}$, the active constraints and feasible normalizations are

$$K(D) = \{k \mid \mathrm{supp}(k) \subseteq D\}, \tag{1}$$

$$\mathcal{N}(D) = \bigcap_{k \in K(D)} S_k. \tag{2}$$

Empty-support logical or ontological constraints are active in every corpus. A constraint derived jointly from several documents is active only while every document in its support remains active.

This fixed ambient universe is deliberate. Documents, claims, constraints, and hypotheses are immutable mathematical objects. Database mutation changes an activation set. A later variable-universe theory may garbage-collect or forget inactive claims through explicit restriction maps, but semantic retraction does not require erasing the objects that explain prior results.

3 The solver is STE

The phrase "constraint solver" can suggest an additional optimization or clustering semantics. There is none here. The solver state is a set $X \subseteq \Omega$. Applying k is literal STE property-set application:

$$\mathrm{apply}(X, k) = X \cap S_k. \tag{3}$$

The exact reduction caused by k is

$$\mathrm{red}(X, k) = X \setminus (X \cap S_k) = \{\omega \in X \mid \omega \notin S_k\}. \tag{4}$$

These are `Model.applyConstraint` and `Model.solverReduction`.

Proposition 3.1 (One-step STE laws)**.** *Property application is contractive and idempotent; any two property applications commute. Moreover,*

$$\mathrm{apply}(X, k) \cup \mathrm{red}(X, k) = X,$$

and $\mathrm{red}(X, k) = \varnothing$ *exactly when* $X \subseteq S_k$*.*

Lean: `applyConstraint_subset`, `applyConstraint_idempotent`, `applyConstraint_comm`, `apply_union_reduction`, and `solverReduction_eq_empty_iff`.

For a finite schedule $[k_1, \ldots, k_n]$, define a run by repeated application, starting from X, and define solving to start from Ω.

Theorem 3.2 (Operational–denotational correspondence)**.** *For every finite list L,*

$$\omega \in \mathrm{run}(L, X) \iff \omega \in X \ \wedge\ \forall k \in L,\ \omega \in S_k, \tag{5}$$

$$\mathrm{solve}(L) = \bigcap_{k \in L} S_k. \tag{6}$$

Consequently, schedules with the same constraint membership have equal output; in particular, permutations and duplicates do not matter.

Lean: `mem_run`, `run_eq_inter_partial`, `solve_eq_partialFeasibilitySet`, `solve_congr`, and `solve_perm`.

If a list enumerates exactly $K(D)$, the final operational state equals $\mathcal{N}(D)$ (`solve_eq_feasible`). Thus a SAT, SMT, ASP, BDD, or custom partition engine is correct only insofar as it implements this meet and its queries. Such technologies are representations and acceleration strategies, not alternate semantics.

Retraction follows the same point. The system does not attempt to invert an old merge. It changes $K(D)$ using provenance and evaluates the new meet. An incremental implementation may reuse caches or truth-maintenance labels, but its extensional correctness criterion is equality with the fresh STE meet. This connects directly to truth-maintenance systems [2, 3], provenance [4], and incremental view maintenance [1].

4 Dynamic corpus laws

Let $D \subseteq E$. Every constraint active in D is active in E, hence

Theorem 4.1 (Information antitonicity)**.**

$$D \subseteq E \implies \mathcal{N}(E) \subseteq \mathcal{N}(D).$$

Lean: `Model.feasible_antitone`. Read forward, document addition removes hypotheses. Read backward, removal can restore hypotheses. The latter is `feasible_subset_after_remove`.

The elementary database laws are exact:

- duplicate insertion is idempotent (`feasible_insert_idempotent`);
- insertion order is irrelevant (`feasible_insert_comm`);
- if $d \notin D$, adding and then removing d restores precisely $\mathcal{N}(D)$ (`feasible_insert_then_remove`).

Define the constraint delta

$$\Delta K(D, E) = K(E) \setminus K(D).$$

Then addition has an exact incremental decomposition.

Theorem 4.2 (Delta decomposition)**.** *If $D \subseteq E$, then*

$$\omega \in \mathcal{N}(E) \iff \omega \in \mathcal{N}(D) \;\wedge\; \forall k \in \Delta K(D, E),\; \omega \in S_k.$$

Lean: `mem_feasible_extension`. If the delta is empty, the feasible sets are equal (`feasible_eq_of_no_new_constraints`).

Constraint activation interacts asymmetrically with union and intersection:

$$K(D \cap E) = K(D) \cap K(E), \qquad K(D) \cup K(E) \subseteq K(D \cup E).$$

The second inclusion can be strict because $D \cup E$ may activate a constraint whose support draws documents from both sides. Lean: `activeConstraints_inter` and `activeConstraints_union_subset`. This is the first genuinely cross-document effect in the theory.

5 Must, may, cannot, and uncertainty

For $c, d \in \mathsf{C}$ and a consistent corpus, define

$$\mathsf{Must}_D(c, d) \iff \forall \omega \in \mathcal{N}(D),\; c \sim_\omega d, \tag{7}$$

$$\mathsf{May}_D(c, d) \iff \exists \omega \in \mathcal{N}(D),\; c \sim_\omega d, \tag{8}$$

$$\mathsf{Cannot}_D(c, d) \iff \forall \omega \in \mathcal{N}(D),\; c \not\sim_\omega d, \tag{9}$$

$$\mathsf{U}_D(c, d) \iff (\exists \omega \in \mathcal{N}(D),\; c \sim_\omega d) \wedge (\exists \omega \in \mathcal{N}(D),\; c \not\sim_\omega d). \tag{10}$$

This is the certain/possible distinction of incomplete-database possible worlds [6] applied to normalization partitions.

Consistency must be checked first. If $\mathcal{N}(D) = \varnothing$, both universal statements are vacuously true. The implemented status type therefore has four constructors:

$$\mathsf{inconsistent}, \quad \mathsf{must}, \quad \mathsf{cannot}, \quad \mathsf{uncertain}.$$

Lean: `Model.CorefStatus` and `Model.corefStatus`, with one correctness theorem for each constructor.

Theorem 5.1 (Safe quotient). Must_D *is an equivalence relation. Therefore the deterministic normalized frame collection*

$$F'_D = \mathsf{C}(D)/\mathsf{Must}_D$$

is well defined on active claims.

Proof. Reflexivity, symmetry, and transitivity hold in every exact relation $\sim_\omega$ and are preserved by intersection over $\omega \in \mathcal{N}(D)$. $\square$

Lean: `mustSame_equivalence`, `activeMustSetoid`, and `CanonicalFrame`.

Possible coreference is reflexive on a consistent corpus and symmetric, but no general transitivity theorem is available. U_D is symmetric and irreflexive. Must and cannot are each disjoint from uncertainty. Lean: `maySame_refl`, `maySame_symm`, `uncertain_symm`, `uncertain_irrefl`, `must_not_uncertain`, and `cannot_not_uncertain`.

5.1 A finite countermodel

Let the mentions be a, b, c and let two exact worlds be feasible. The left world has blocks $\{a, b\}, \{c\}$; the right world has blocks $\{a\}, \{b, c\}$. Each world is an ordinary partition. Nevertheless,

$$\mathsf{May}(a, b), \qquad \mathsf{May}(b, c), \qquad \neg\mathsf{May}(a, c).$$

Moreover (a, b) and (b, c) are uncertain while (a, c) is cannot-corefer.

Theorem 5.2 (Nontransitivity). *Neither* May *nor* U *is transitive in general.*

The model and proofs are fully finite and constructive in `Ste.DynamicFrame.Counterexample`; the theorem names are `maySame_not_transitive` and `uncertain_not_transitive`.

This establishes a design constraint. U is an overlay between must-classes, not a special kind of class. Taking connected components of possible or uncertain edges invents identifications not licensed in any exact world.

6 Restriction, quotient maps, and retraction asymmetry

On the fixed ambient hypothesis universe, inclusion $D \subseteq E$ induces

$$\rho_{E,D} : \mathcal{N}(E) \longrightarrow \mathcal{N}(D), \qquad \rho_{E,D}(\omega) = \omega.$$

This is a genuine restriction map: it forgets the additional requirements, not the underlying hypothesis.

Theorem 6.1 (Restriction laws). *Restriction by identity is identity, and for $D \subseteq E \subseteq F$,*

$$\rho_{E,D} \circ \rho_{F,E} = \rho_{F,D}.$$

A point of $\mathcal{N}(D)$ lies in the image of $\rho_{E,D}$ exactly when the same ambient hypothesis is feasible in E.

Lean: `restrictFeasible_id`, `restrictFeasible_comp`, and `mem_range_restrictFeasible_iff`.

Thus $D \mapsto \mathcal{N}(D)$ is already a contravariant set-valued construction on the poset of active document sets. Calling it a presheaf becomes literal once the corpus-dependent universe and a target category are fixed. Calling it a sheaf would require a further gluing theorem; that is an open problem, not terminology assumed here.

The quotient moves the other way. Since

$$\mathsf{Must}_D(c,d) \Longrightarrow \mathsf{Must}_E(c,d),$$

old must-classes map into new must-classes. Active-claim inclusion and relation preservation induce

$$q_{D,E} : F'_D \longrightarrow F'_E.$$

Lean: `includeActiveClaim` and `canonicalMap`. This map need not be injective: added evidence can eliminate all split worlds and thereby merge old must-classes. It also need not be surjective because E contains new active claims.

There is generally no quotient map in the deletion direction. A pair forced together in E may become uncertain in D after the supporting document is removed. Mapping one E-class back would then require splitting it, which is not a function on quotient classes. This is the formal reason that deletion cannot be implemented as the inverse of an agglomerative merge history.

7 Minimal document supports

An explanation should identify documents sufficient for a settled conclusion. For a finite document set D, say D supports a must-link when $\mathcal{N}(D)$ is nonempty and $\mathsf{Must}_D(c,d)$ holds. Define cannot-support analogously. A support is minimal when no proper subset still supports the conclusion.

Theorem 7.1 (Finite minimal certificates)**.** *If a finite corpus supports a must-coreference conclusion, it contains an inclusion-minimal finite support for that conclusion. The same holds for a cannot-coreference conclusion.*

Proof. Among the finite supporting subsets of the original corpus, choose one of minimum cardinality. Every proper subset has smaller cardinality and therefore cannot support the conclusion. □

Lean: the generic finite minimization theorem is `exists_minimal_subfinset`; its two applications are `exists_minimal_mustSupport` and `exists_minimal_cannotSupport`.

The theorem is semantic, not an efficiency claim. Computing all minimal supports may be exponential. ATMS labels, unsatisfiable cores, provenance polynomials, and hitting-set dualization are candidate representations. The proof does establish a precise contract for an explanation engine: a returned certificate should be sufficient, consistent, and inclusion-minimal.

8 Lifting truth over normalization worlds

Let T be a type of downstream truth worlds and let

$$T : \Omega \longrightarrow \mathcal{P}(\mathsf{T})$$

assign a truth feasible set to each exact normalization. The joint feasible set is

$$J(D) = \{(\omega, t) \mid \omega \in \mathcal{N}(D),\ t \in T(\omega)\}.$$

This is `TruthLift.jointFeasible`. A property $P(\omega, t)$ is certain when it holds throughout $J(D)$ and possible when it holds somewhere in $J(D)$.

Theorem 8.1 (Truth-lift monotonicity). *If $D \subseteq E$, every property certain over $J(D)$ is certain over $J(E)$, and every property possible over $J(E)$ was possible over $J(D)$.*

Lean: `certain_mono` and `possible_antitone`.

Not every feasible normalization need support a downstream truth world. The projection of $J(D)$ onto normalization hypotheses is contained in $\mathcal{N}(D)$, with equality if every feasible normalization has a nonempty truth fiber. Lean: `normalizationProjection_subset` and `normalizationProjection_eq`.

The first commutation result covers the clean independence case.

Theorem 8.2 (Normalization-independent commutation). *Assume $\mathcal{N}(D) \neq \varnothing$, $T(\omega) = T_0$ for every ω, and $P(\omega, t) = P_0(t)$. Then*

$$\forall(\omega, t) \in J(D),\ P_0(t) \iff \forall t \in T_0,\ P_0(t).$$

Existentially, joint possibility is equivalent to consistency of $\mathcal{N}(D)$ together with ordinary possibility in T_0.

Lean: `certain_constant_iff` and `possible_constant_iff`.

This identifies the exact modularity boundary. Normalization-first inference commutes automatically when truth semantics ignores normalization. The research problem is to weaken that assumption: characterize dependent $T(\omega)$ and $P(\omega, t)$ for which certain answers can be computed without enumerating every joint world.

9 Status evolution under corpus change

The modal laws give a concise transition discipline. Under $D \subseteq E$:

$$\begin{aligned}
\mathsf{Must}_D(c, d) &\Longrightarrow \mathsf{Must}_E(c, d), \\
\mathsf{Cannot}_D(c, d) &\Longrightarrow \mathsf{Cannot}_E(c, d), \\
\mathsf{May}_E(c, d) &\Longrightarrow \mathsf{May}_D(c, d), \\
\mathsf{U}_E(c, d) &\Longrightarrow \mathsf{U}_D(c, d).
\end{aligned}$$

Thus addition can resolve uncertainty into must or cannot, or make the entire problem inconsistent. It cannot turn a consistent settled fact into uncertainty within the hard-constraint semantics. Deletion reverses the information order: a settled link may become uncertain, but a possibility cannot be invented by adding a constraint.

These are `mustSame_mono`, `cannotSame_mono`, `maySame_antitone`, and `uncertain_antitone`. They are useful both as theorems and as property-based tests for an incremental implementation.

10 Relationship to learned coreference

Nothing in the theory forbids neural models. They can propose candidate frame interpretations, candidate links, and likely high-reduction criteria. They can rank feasible worlds. What they cannot do silently is turn an uncalibrated score threshold into a semantic impossibility.

This yields three distinct interfaces:

1. *generation*: place defensible worlds inside the ambient candidate universe;

2. *hard admissibility*: apply audited property sets by STE;

3. *ranking*: order the worlds that remain.

Candidate recall is a fairness condition on generation. Hard-constraint soundness is a fairness condition on property sets. Ranking does not alter must/cannot guarantees unless the ranked tail is explicitly discarded, in which case that discard is itself a property-set application that must be named and audited.

11 Mechanization map

The development is split by responsibility:

Lean module	Formal content
`Ste.DynamicFrame`	model, feasibility, modal relations, quotient
`Ste.DynamicFrame.Laws`	dynamic laws, restriction, status, quotient map
`Ste.DynamicFrame.Solver`	property application, reduction, finite runs
`Ste.DynamicFrame.Counterexample`	three-mention nontransitivity model
`Ste.DynamicFrame.Support`	finite minimal document certificates
`Ste.DynamicFrame.TruthLift`	dependent downstream feasible family

The root module `Ste.lean` imports the complete theory. The project is pinned to Lean 4.32.0 and Mathlib 4.32.0. The repository workflow builds the root library on pushes to the development branch. The source and this paper use a strict convention: a named Lean theorem is the authoritative statement; claims listed below as conjectures are not represented as proved facts.

12 Open theory

The present development closes several elementary questions but opens a more focused research program.

Conjecture 12.1 (Variable-universe presheaf). There is a category of corpus-dependent exact normalizations whose restriction maps forget inactive claim occurrences and satisfy identity and composition, and whose fixed-universe embedding recovers `restrictFeasible`.

Conjecture 12.2 (Gluing classification). Compatible local normalizations admit a global extension exactly when a computable obstruction vanishes. Uniqueness of gluing corresponds to absence of residual cross-document identity ambiguity.

Conjecture 12.3 (Compact exact solver). For practically generated candidate graphs, the feasible family of partitions admits a compact representation supporting document insertion, deletion, must/may queries, and minimal-support extraction without full model enumeration.

Conjecture 12.4 (Typed relation algebra). Strict identity, subevent, supersession, temporal order, causation, repeated instance, and framing divergence admit a sound typed composition algebra whose necessary consequences are preserved by feasible-world intersection.

Conjecture 12.5 (Dependent truth commutation). There are checkable conditions weaker than constant truth fibers under which normalization-first certain answers equal the certain answers of a fully joint normalization/truth solver.

Conjecture 12.6 (Stable presentation identity). Canonical-frame presentation identifiers can be maintained with a precise lineage semantics through merges, splits, and re-merges, without treating any historical point estimate as authoritative identity.

Other engineering questions now have exact correctness oracles: incremental output must equal a fresh STE meet; insertion order must not matter; deletion must reactivate every world excluded only by removed support; and every published settled link should admit a finite minimal certificate when the active corpus is finite.

13 Conclusion

Dynamic frame normalization is best understood as set-theoretic estimation over exact partitions, not as an irreversible clustering task. Property-set application is the solver. Provenance selects the active meet. Must coreference is the intersection of feasible equivalence relations and supplies the only generally safe deterministic quotient. 'May' and U are modal summaries, and a finite machine-checked countermodel shows why they cannot be collapsed into partitions. Document addition and deletion act in opposite information directions; feasible worlds restrict contravariantly, while canonical quotients map covariantly under addition. Minimal finite supports provide an explanation contract, and the dependent truth lift keeps normalization uncertainty intact downstream.

The result is a small but substantive mathematical kernel for a live document database: ambiguity is represented, reduction is exact, retraction has a denotation, and the remaining research questions are stated at the level where they can be proved or refuted.

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